EMATH 100

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Introduction to Radicals

Simplifying Radicals

Adding and Subtracting Radicals

Multiplying and Dividing Radicals

Solving Equations Containing Radicals Radical Equations and Problem Solving

Introduction to Radicals

Square Roots

Opposite of squaring a number is taking the *square root* of a number.

A number *b* is a square root of a number *a* if $b^2 = a$.

In order to find a square root of *a*, you need a # that, when squared, equals *a.*

Principal Square Roots

The *principal* (*positive*) *square root* is noted as

$\sqrt{a}$

The *negative square root* is noted

as $-\sqrt{a}$

Radicands

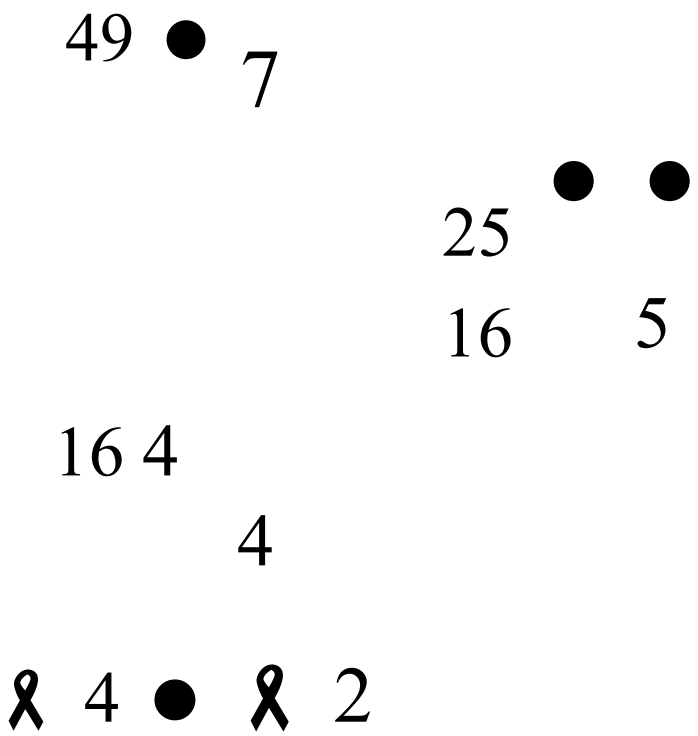
Radical expression is an expression containing a *radical sign*.

Radicand is the expression under a radical sign.

Note that if the radicand of a square root is a negative number, the radical is NOT a real number.

Radicands

Example



Perfect Squares

Square roots of *perfect square*

radicands simplify to rational numbers (numbers that can be written as a quotient of integers).

Square roots of numbers that are not perfect squares (like 7, 10, etc.) are *irrational numbers*.

IF REQUESTED, you can find a decimal *approximation* for these irrational numbers.

Otherwise, leave them in radical form.

Perfect Square Roots

Radicands might also contain variables and powers of variables.

To avoid negative radicands, assume for this chapter that if a variable appears in the radicand, it represents positive numbers only.

Example

$\sqrt[10]{64x^5}\sqrt[5]{8x}$

- Cube Roots

The *cube root* of a real number a $a \bullet b\, b \bullet a^{3}{}^{3}$ only if

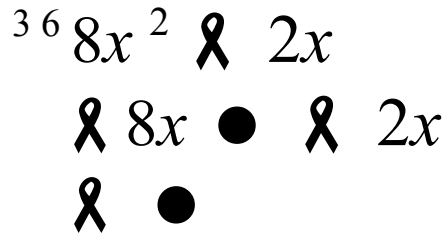
Note: a is not restricted to non negative numbers for cubes.

Cube Roots

Example

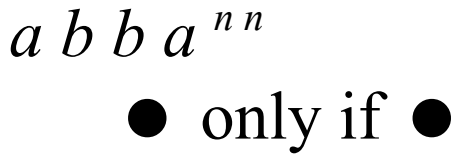
$\sqrt[3]{27} = 3$

$\bullet$



*n*th Roots

Other roots can be found, as well. The *nth root* of *a* is defined as



If the *index*, *n*, is even, the root is

NOT a real number when *a* is negative.

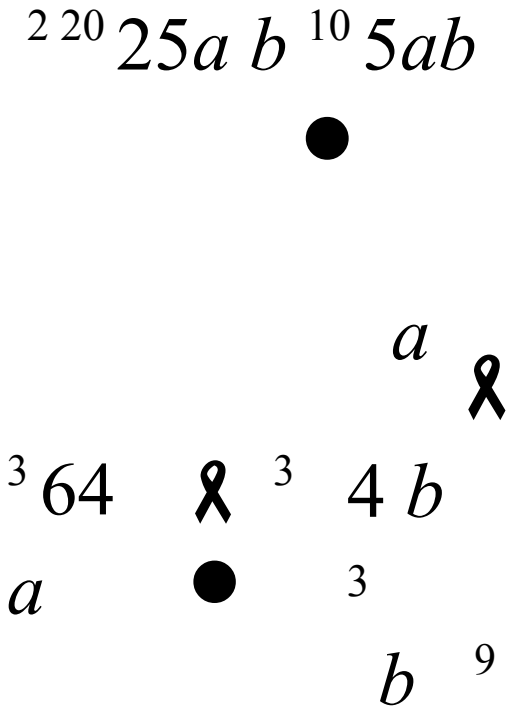
If the index is odd, the root will be a real number.

Example

*n*th Roots



Simplify the following.



Simplifying Radicals

Product Rule for Radicals

If and are real numbers, $a\ b$

ab ● a 🌥 b

ab ● if b

a 📻 $0\ b$

Simplifying Radicals

⚠ Example

Simplify the following radical expressions. $40 \cdot 4 \quad 10 \cdot 2\,10$

5

$5 \cdot 5$

$16 \cdot$

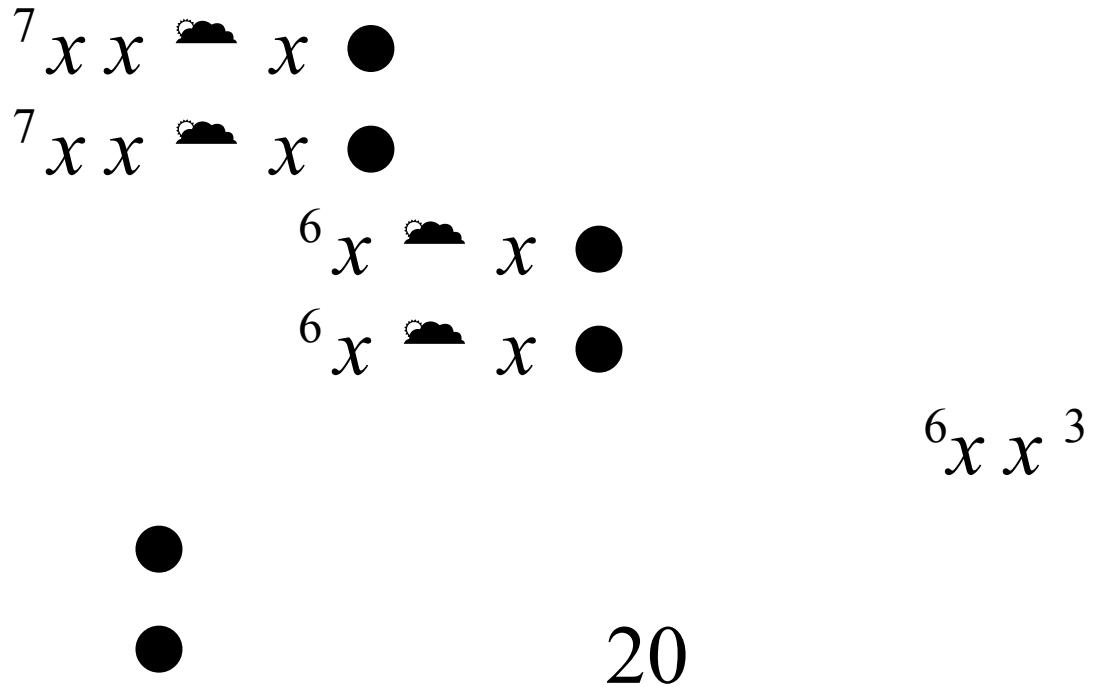
$16 \quad 4$

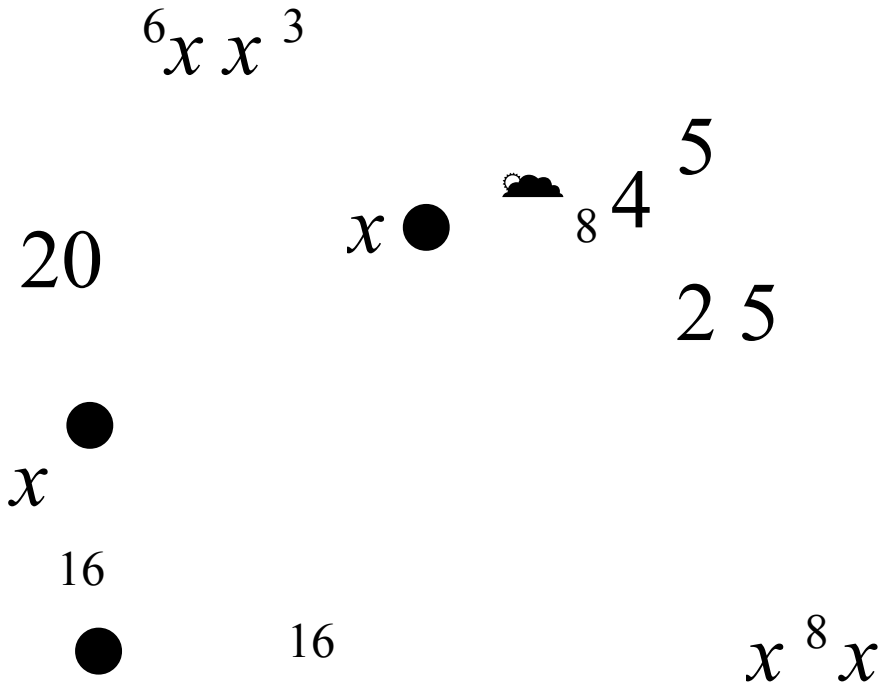
15 No perfect square factor, so the radical is already simplified.

Simplifying Radicals

Example

Simplify the following radical expressions.





Quotient Rule for Radicals

$^{n}a^{n}$ If and are real numbers,

$$b^{nnn}ab \bullet a \text{ } b$$

$$a^{n} \quad a$$

$$^{n}b$$

$$b \qquad 0^{n}$$

$$\bullet \text{ if }$$

^{n}

$$b$$

Simplifying Radicals

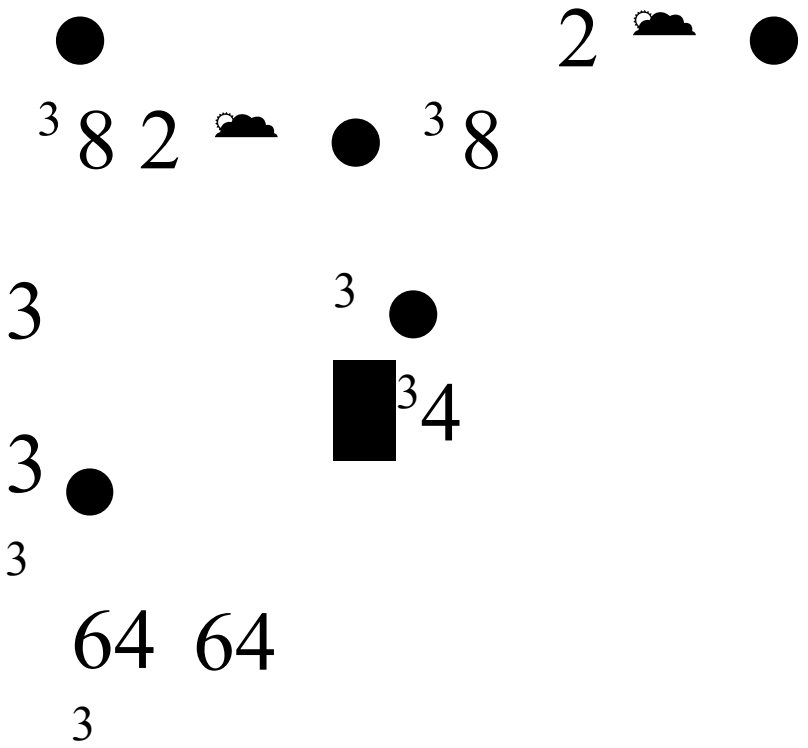
⚠ **Example**

Simplify the following radical expressions. $\sqrt[3]{8}$ $\sqrt[3]{2}$

$2\ 2$

$\sqrt[3]{16}$ $\sqrt[3]{\ }$ 16 $\sqrt{\ }$

$\sqrt[3]{8}\ \sqrt[3]{2}\ 2\ 2$



Adding and Subtracting Radicals

Sums and Differences

Rules in the previous section allowed us

to split radicals that had a radicand which was a product or a quotient.

We can NOT split sums or

differences. $a \pm b$

$\sqrt{a \pm b}$

$a \ \text{Ribbon} \ b \neq a \ \text{Ribbon} \ b$

Like Radicals

In previous chapters, we've discussed the concept of "like" terms.

These are terms with the same variables raised to the same powers.

They can be combined through addition and subtraction.

Similarly, we can work with the concept of "like" radicals to combine radicals with the same radicand.

Like radicals are radicals with the same index and the same radicand.

Like radicals can also be combined with addition or subtraction by using the distributive property.

Adding and Subtracting Radical Expressions

⚠ **Example**

$3\sqrt{7}\ 3 \bullet 8\ 3$

$10\ 2\ \text{ribbon}\ 4\ 2 \bullet 6\ 2$

$\sqrt[3]{2}\ 4\ 2$ Can not simplify

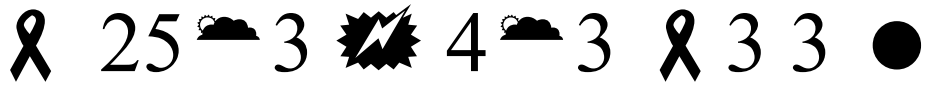
$\blacksquare\sqrt{\ }\blacksquare$ Can not simplify

Adding and Subtracting Radical Expressions

Example

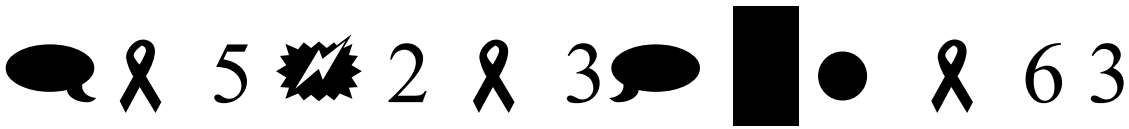
Simplify the following radical expression.







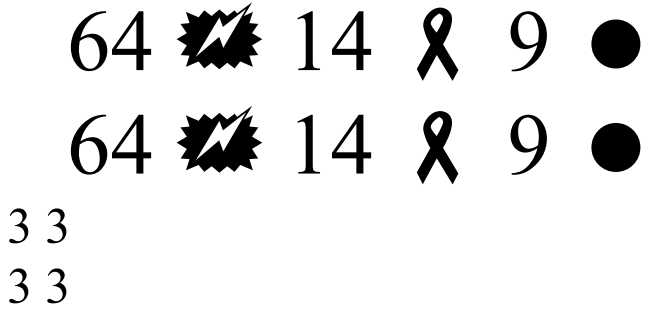


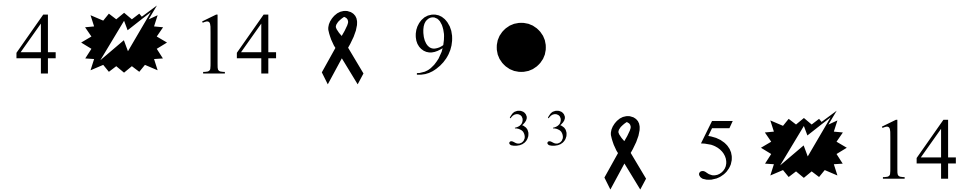


Adding and Subtracting Radical Expressions

⚠ **Example**

Simplify the following radical expression.





Expressions



Example

Simplify the following radical expression. Assume that variables represent positive real numbers.

$$\sqrt[3]{3\ 9x} \cdot 5x \cdot x\ 5x \bullet$$

$$\sqrt[3]{3\ 9x} \cdot 5x \cdot x\ 5x \bullet$$

$$2$$

$$2$$

$$\sqrt{3\ 45x} \cdot x\ 5x \bullet$$

$$\sqrt{3\ 45x} \cdot x\ 5x \bullet$$

$3\ 9x \cdot 5x \cdot x\ 5x \bullet$

$3 \cdot 3x\ 5x \cdot x\ 5x \bullet$

$9x\ 5x \cdot x\ 5x \bullet$

$9x \cdot x\ 5x \bullet\ 10x\ 5x$

Multiplying and Dividing Radicals

Multiplying and Dividing Radical Expressions

$\sqrt[n]{a}\ \sqrt[n]{}$ If and [redacted] are real numbers,

$\sqrt[n]{a} \cdot \sqrt[n]{b} \bullet \sqrt[n]{ab}$

$$\sqrt[n]{a^n} = a$$

- $\sqrt[n]{b}$ if $0 \leq b$
- $\sqrt[n]{b}$ $b \geq 0$

Multiplying and Dividing Radical Expressions

⚠ **Example**

Simplify the following radical expressions. $\sqrt{3y} \cdot \sqrt{5x} = \sqrt{15xy}$

$\sqrt{3y} \cdot \sqrt{5x} = \sqrt{15xy}$

$a^{7}b^{6}$

$a^{7}b^{6}$

$a^{3}b^{2}$

$a^{3}b^{2}$

$a^{4}b^{4}a^{2}b^{2}a$

b

Rationalizing the Denominator

Many times it is helpful to rewrite a radical quotient with the radical confined to ONLY the

numerator.

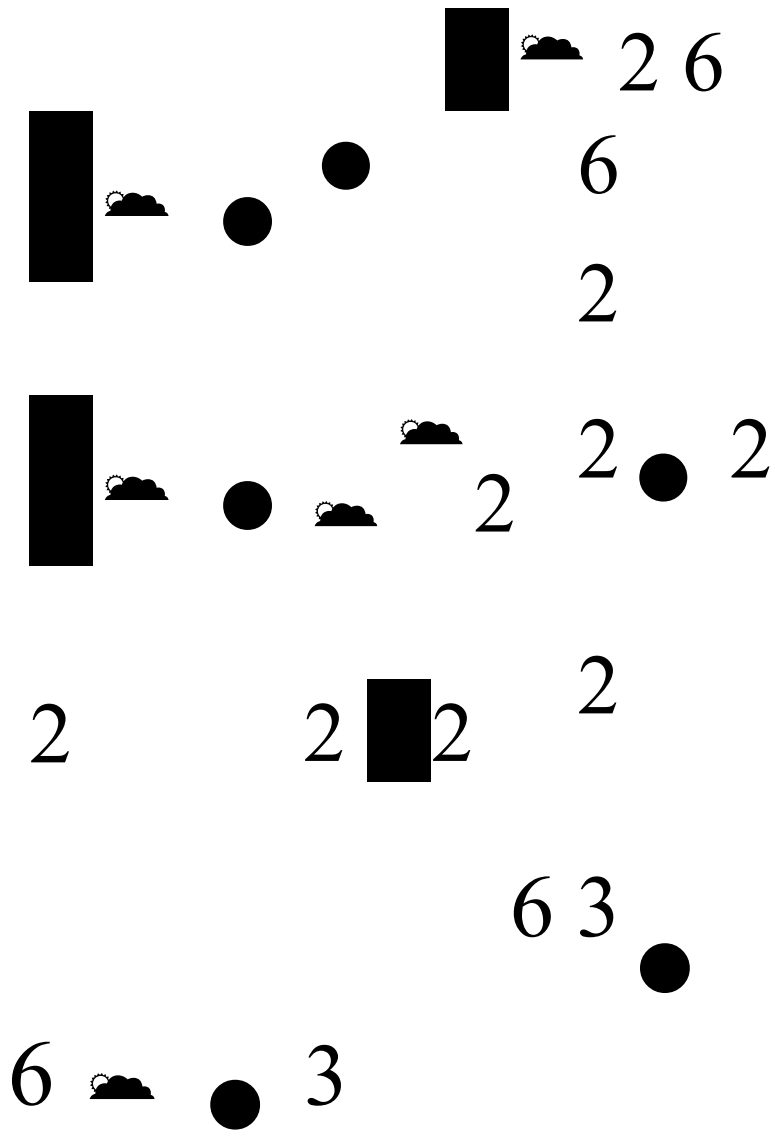
If we rewrite the expression so that there is no radical in the denominator, it is called *rationalizing the denominator*.

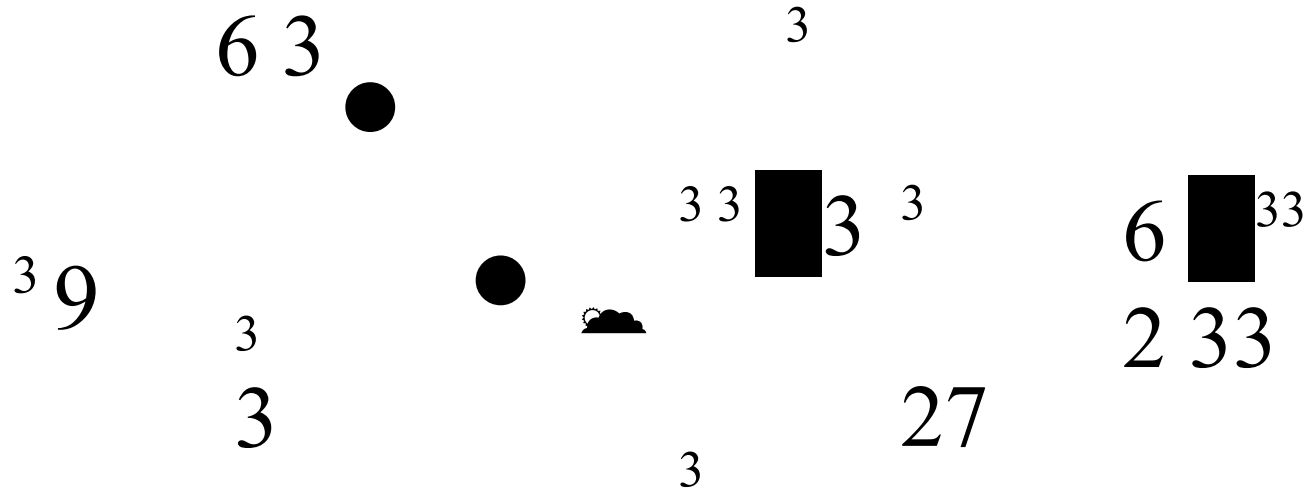
This process involves multiplying the quotient by a form of 1 that will eliminate the radical in the denominator.

Rationalizing the Denominator



Rationalize the denominator.


2 6
6
2
2
2 2
2
2
2
2
2
6 3
6
3



Conjugates

Many rational quotients have a sum or difference of terms in a denominator, rather than a single radical.

In that case, we need to multiply by the *conjugate* of the numerator or

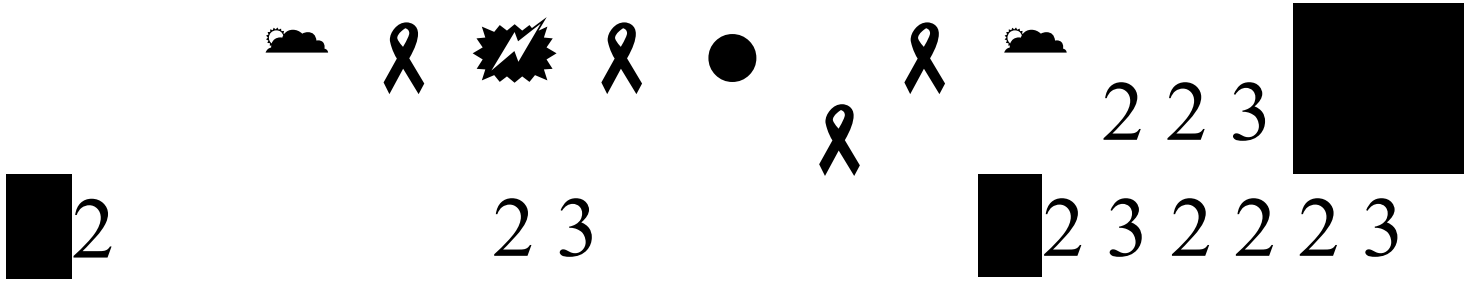
denominator (which ever one we are rationalizing).

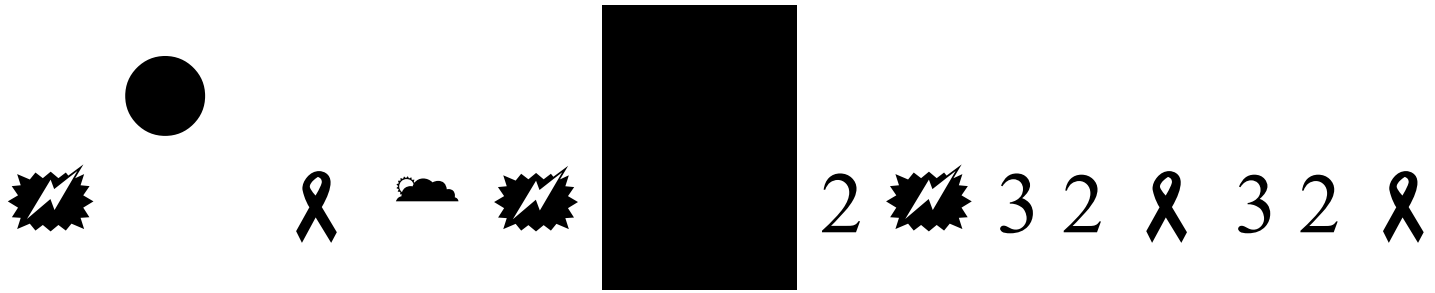
The conjugate uses the same terms, but the opposite operation (+ or).

Rationalizing the Denominator

Example

Rat ominator.



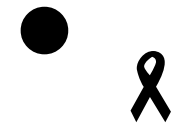


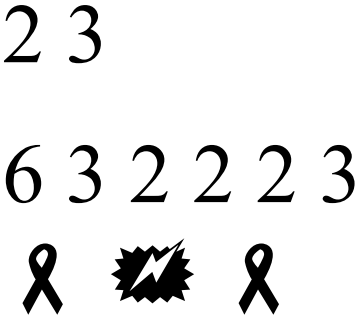


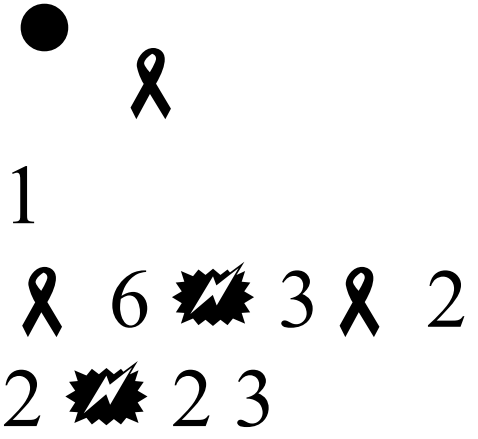












Solving Equations Containing Radicals

Extraneous Solutions

Power Rule (text only talks about squaring, but applies to other powers, as well).

> If both sides of an equation are raised to the same power, solutions of the new equation contain all the solutions of the original equation, but might also contain additional solutions.

A proposed solution of the new equation that is NOT a solution of the original

equation is an *extraneous solution*.

Solving Radical Equations

Example

Solve the following radical equation.

$x - 1 = 5$ Substitute into the original equation.

$(\quad)^2 \quad 2x - 1 = x - 24$ original equation.

5

$x - 1 = 25$

$24 = 5$

$25 \stackrel{?}{=} 5$ *true*

So the solution is $x = 24$.

Solving Radical Equations

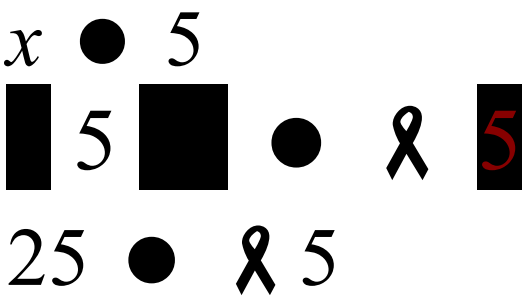
Example

Solve the following radical equation. $\sqrt{5x} - 5 = 0$

$\sqrt{5x} \stackrel{?}{=} 5$ Substitute into the original equation.

$(\sqrt{5x})^2 \stackrel{?}{=} 5^2$

$5x \stackrel{?}{=} 25$



the left side of the equation is asking for the principal square root.

Does NOT check, since

So the solution is .

Solving Radical Equations

Steps for Solving Radical Equations 1) Isolate one radical on one side of equal sign. 2) Raise each side of the equation to a power

equal to the index of the isolated radical,

and simplify. (With square roots, the index is 2, so square both sides.)

3) If equation still contains a radical, repeat steps 1 and 2. If not, solve equation.

4) Check proposed solutions in the original equation.

Solving Radical Equations

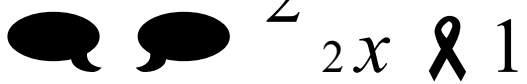
⚠ Example

Solve the following radical equation.

$x - 1 - 1 = 0\,x$

$- 1 = 1$

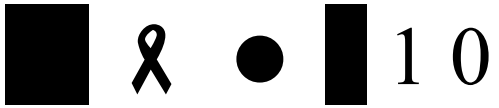
$x - 1 = 1$



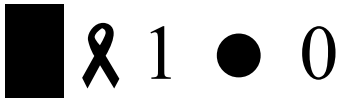
original equation.

original equation. 2













Substitute into the

So the solution is $x = 2$.

Solving Radical Equations Example

Solve the following radical

equation. $2x - x - 1 = 8$

$x - 1 = 8 + 2x$

$x - 1 = 8 + 2x$

$(\quad)^2$

$x - 1 = 8 + 2x$

$$x - 1 = 64 + 32x - 4x^2$$

$$0 = 63 + 33x - 4x^2$$

$$0 = (3 + x)(21 + 4x)$$

$$x = -3 \text{ or } x = -\frac{21}{4}$$

Solving Radical Equations

Example continued

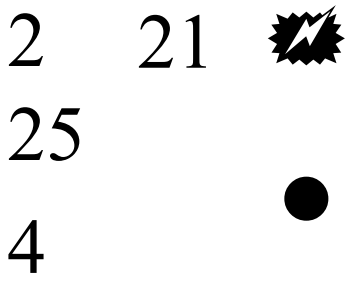
Substitute the value for *x* into the original equation, to check the solution.











5

$8\ 8$

$2\ \ 2$

26 *false* So the solution is $x = 3$.

$2\ \ 8$

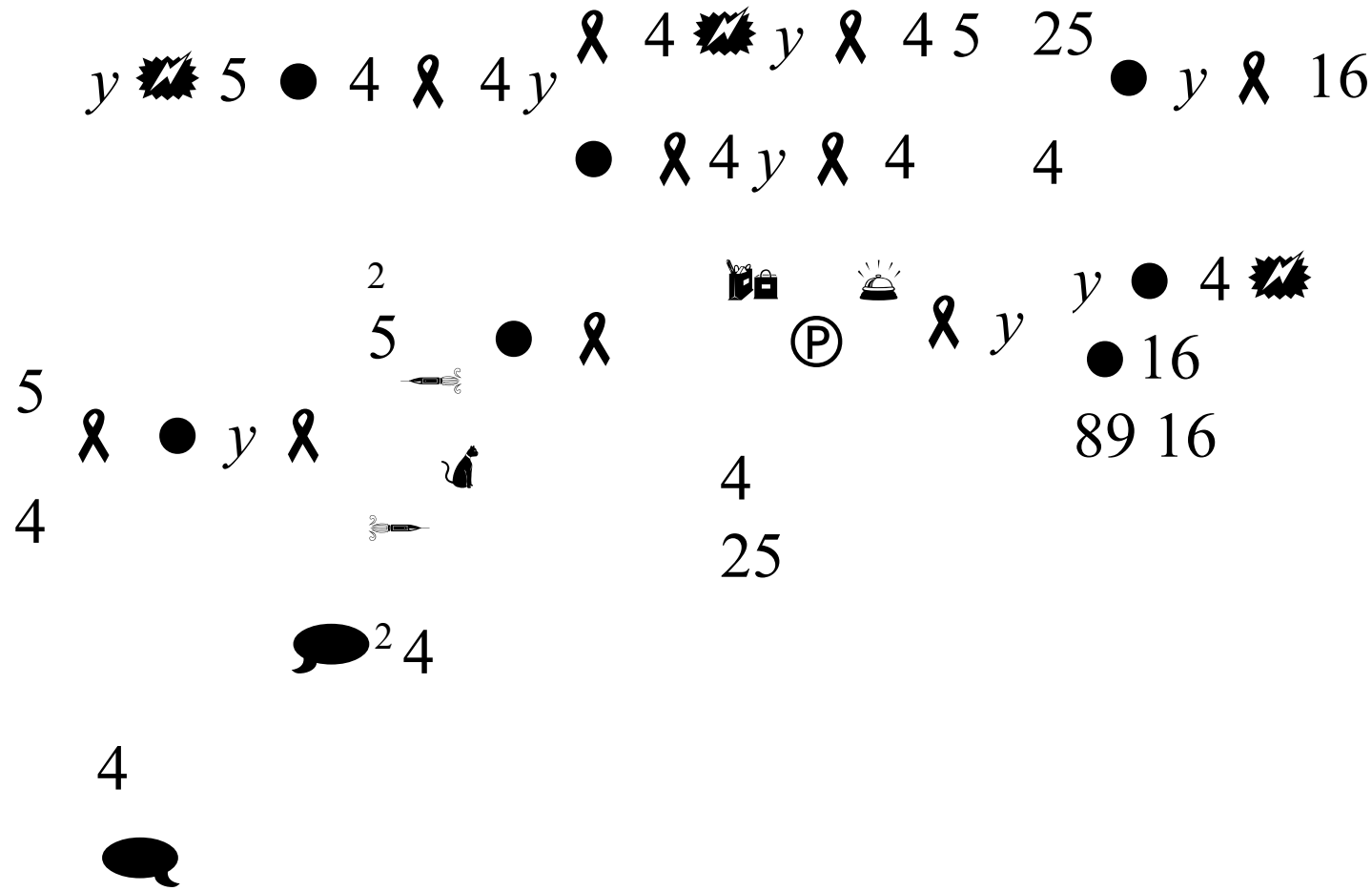
Solving Radical Equations Example

Solve the following radical equation.

$y - 5 = 2 - y - 4$

$(y - 5)^2 = (2 - y - 4)^2$

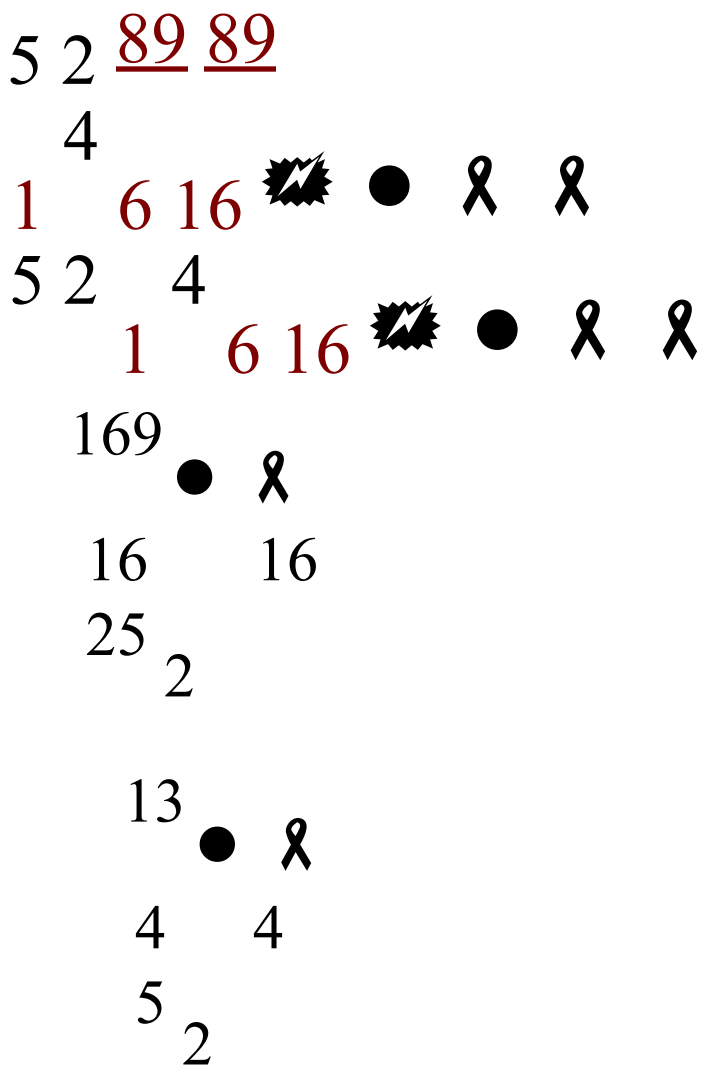
$y - 5 = 2 - y - 4$



Solving Radical Equations

Example continued

Substitute the value for *x* into the original equation, to check the solution.



13 ● *false* So the solution is .3

4 4

Solving Radical Equations Example

Solve the following radical equation. $2x - 4 - 3x = 4 \bullet -2$

$2x - 4 \bullet -2 = 3x = 4$

$(\quad)^2 = (\quad)^2$

$(\quad)^2 = (\quad)^2$

$2x - 4 \bullet -2 = 3x = 4$

$2x - 4 \bullet -2 = 3x = 4$

$$2x \;\square\; 4 \;\bullet\; 4 \;\square\; 4\,3x \;\square\; 4 \;\square\; 3x \;\square\; 4$$

$$2x \;\square\; 4 \;\square\; 8 \;\square\; 3x \;\bullet\; \square\; 4\,3x \;\square\; x \;\square\; \bullet\; x \;\square\; \bullet\; x \;\square\; \square\; 4$$

$$\square\; x \;\square\; 12 \;\bullet\; \square\; 4\,3x \;\square\; 4$$

$$\square\; \square\; \square\; \square^{2\,2} \;\square\; x \;\square\; 12 \;\bullet\; \square\; 4\,3x \;\square\; 4$$

$$24\;144\;16(3\;4)\;48\;64 \;\square^{2}\; x \;\square$$

$$24\;80\;0 \;\square^{2}$$

$$x \;\square\; x \;\square\; \bullet\; \square x \;\square$$

$$20\square\square x \;\square\; 4\square \;\bullet\; 0\,x \;\bullet$$

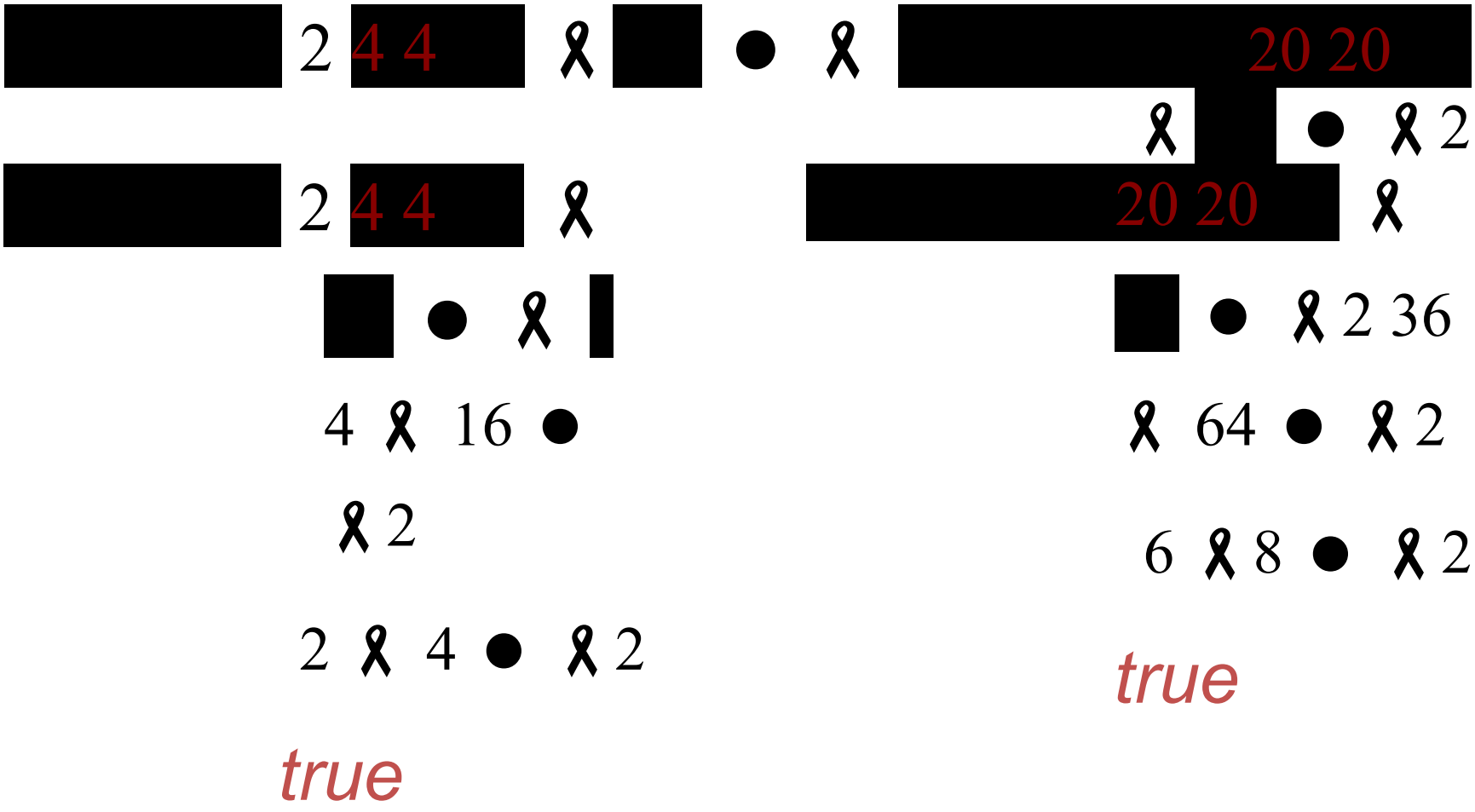
$$4 \text{ or } 20$$

Solving Radical Equations

⚠

Example continued

Substitute the value for *x* into the original equation, to check the solution.



So the solution is $x = 4$ or 20.

Radical Equations and Problem Solving

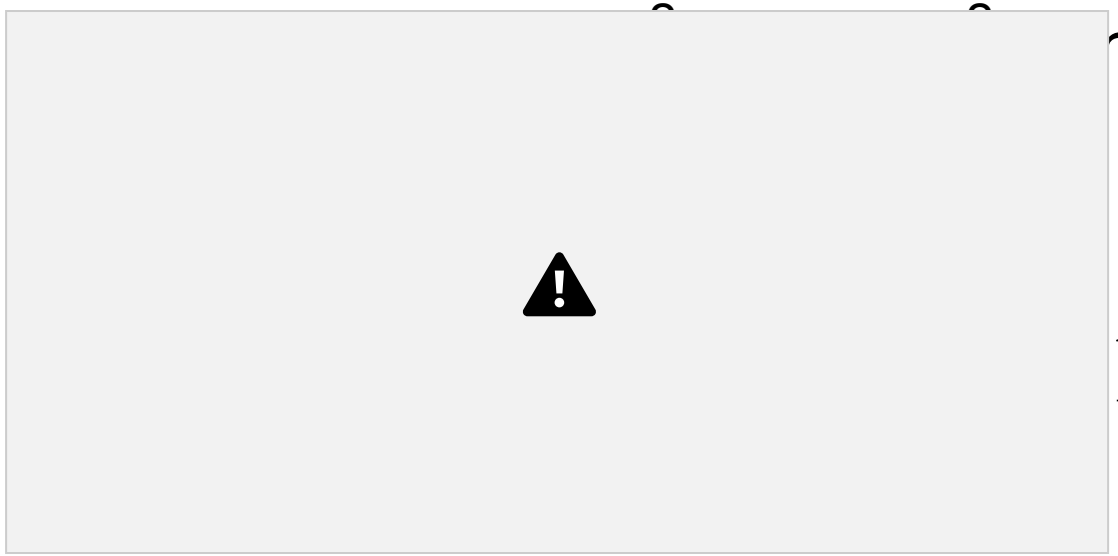
The Pythagorean Theorem

Pythagorean Theorem

In a right triangle, the sum of the squares of the lengths of the two legs is equal to the square of the length of the hypotenuse.

$$(\text{leg } a)^2 + (\text{leg } b)^2 = (\text{hypotenuse})^2$$

$$\text{hypotenuse})^2$$



leg a hypotenuse

leg b

Using the Pythagorean Theorem

Example

Find the length of the hypotenuse of a right triangle when the length of the two legs are 2 inches and 7 inches.

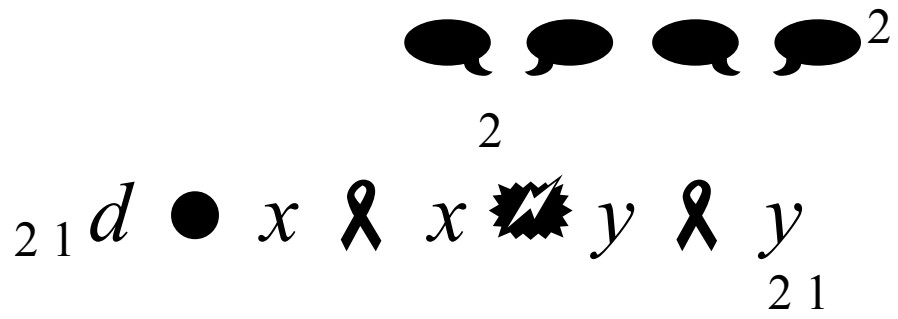
$$c^2 = 2^2 + 7^2 = 4 + 49 = 53$$

c = inches $\sqrt{53}$

The Distance Formula

By using the Pythagorean Theorem, we can derive a formula for finding the distance between two points with coordinates (x_1,y_1) and (x_2,y_2).

and (x_2,y_2).



The Distance Formula

Example

Find the distance between (5, 8) and (2, 2).



